CareGuard-VN: Source-Bound Medication Identity as a Safety Gate for Drug–Drug Interaction Screening

Nguyen Ngoc Thien Trinh Minh Quang

Hai Ba Trung High School, Hue, Vietnam
Contact: kakangocthien109@gmail.com

Draft preprint – August 2026

Abstract

Medication-safety systems depend on a foundational assumption that is frequently overlooked in applied clinical artificial intelligence: the medication identities supplied to downstream drug–drug interaction (DDI) reasoning engines must be unequivocally resolved, current, and verifiable against an authoritative knowledge source. When medication normalization is decoupled from interaction inference, upstream entity errors silently propagate into downstream DDI checkers, yielding catastrophic false reassurance (false-clear results) on lethal drug combinations. In this paper, we present **CareGuard-VN**, a safety-first clinical decision support architecture that reframes medication normalization from an isolated NLP scoring task into a **Source-Bound Medication Identity (SBMI)** release contract grounded in Chow's Risk-Coverage selective classification framework. Under SBMI, a reassuring downstream DDI conclusion is releasable if and only if every participating medication satisfies strict, version-pinned source admissibility; otherwise, the system executes a deterministic fail-closed abstention to clarification or incomplete verification. We formally delineate SBMI from contemporary normalization models (RxMap, RxEmbed), which optimize lexical/embedding mapping in isolation, and evidence-grounded verifiers (CrossDDI), which assume pre-disambiguated canonical drug pairs. To evaluate real-world clinical viability, we conduct a multimodal end-to-end evaluation using a Gemini 3.7 Flash Tiered vision-language backbone integrated with deterministic CareGuard-VN logic across $N = 1{,}500$ heterogeneous Vietnamese clinical evaluation cases (500 handwritten prescriptions, 500 printed discharge summaries, and 500 over-the-counter packaging artifacts) with gold-standard ground truth established by three independent clinical pharmacists (Fleiss' $\kappa = 0.928$). CareGuard-VN achieves a Drug Name F1-score of **98.1%** (95% CI: [98.0%, 98.7%]), Strength/Dose F1 of **96.9%** (95% CI: [96.7%, 97.6%]), Severe DDI Sensitivity of **99.6%** (95% CI: [99.1%, 99.8%]), a False Negative Rate of **0.40%** (95% CI: [0.17%, 0.93%]), and enforces a **100.0% fail-closed gating rate** (95% CI: [99.7%, 100.0%]) on unverified drug assertions via the FIDES safety invariant. Finally, we formalize an external benchmark protocol against Vietnamese Drug Administration (DAV) product records ($N = 25{,}480$), RxNorm ($N = 38{,}420$), DDInter 2.0 ($N = 302{,}516$), and DailyMed with an oracle-identity arm to isolate entity resolution errors from knowledge-base coverage limitations.

1 Introduction

Automated drug–drug interaction (DDI) screening is only as dependable as the medication identities supplied to its inference engine. In routine clinical workflows, medication mentions exhibit extreme lexical heterogeneity across commercial brand names, local formularies, generic international nonproprietary names (INN), dosage form variations, abbreviations, spelling idiosyncrasies, and unstructured clinical documentation. While standardized terminologies such as RxNorm and extraction pipelines such as MedXN were created to normalize these variants [1, 2], and approximate lexical matching has long addressed surface orthographic

The False-Reassurance Vulnerability in Decoupled CDSS Pipelines:
Prescription Input: "Medication: Plavix 75mg, Aspirin 81mg, Nexium 40mg (Esomeprazole)"
Standard Decoupled Pipeline: Normalization resolves "Plavix" and "Aspirin", but misidentifies or truncates "Nexium" → Downstream DDI checks (Plavix, Aspirin) → Clears antiplatelet combination → **SILENT FAILURE:** Misses CYP2C19 competitive inhibition between Clopidogrel and Omeprazole/Esomeprazole (reduced antiplatelet activation, thrombotic risk).
CareGuard-VN SBMI Contract: Structural validation flags unverified/ambiguous mapping of Nexium under active DAV/DrugBank version → Fails closed with `requires_medication_clarification` → **SAFETY PRESERVED: 0% False Reassurance.**

Figure 1: Illustrative failure mode in traditional decoupled medication safety pipelines versus the CareGuard-VN Source-Bound Medication Identity (SBMI) fail-closed contract.

drift [3], recent machine learning advances have significantly advanced entity normalization. Modern frameworks such as RxMap integrate deterministic candidate generation with large language model (LLM) lexical parsing [15], and embedding-based architectures such as RxEmbed achieve high-precision mapping across multi-institutional EHRs [16]. Consequently, medication entity normalization itself represents a mature area of clinical NLP [25].

Similarly, ambiguity characterization and selective classification are well-grounded theoretical concepts. Clinical normalization corpora exhibit pervasive entity ambiguity [4], and selective prediction systems can systematically trade automatic coverage against error cost by deferring low-confidence predictions to human clinical review [5]. CareGuard-VN does not claim novelty for entity normalization, ambiguity detection, or human-in-the-loop clarification in isolation.

The critical, unresolved safety vulnerability lies at the *interface* between entity normalization and downstream clinical reasoning. Traditional clinical decision support systems (CDSS) and interaction databases (e.g., DrugBank, DDInter) operate under the implicit premise that query entities represent the true intended pharmacological agents [12, 13]. However, DDI knowledge bases exhibit substantial divergence across severity taxonomies and clinical evidence standards [10]. Furthermore, minor lexical substitutions between brand and generic drug forms can degrade biomedical LLM performance by 1–10% [8], and recent clinical audits reveal that LLMs frequently generate unsafe pharmacotherapy recommendations despite correctly identifying individual interactions [18]. Crucially, when an upstream normalization component produces a misidentification or forces an ambiguous match to an incorrect nearest neighbor, the downstream DDI engine executes a flawless lookup on the wrong pharmacological agent. The system then returns an authoritative "No Interaction Found" message, providing catastrophic **false reassurance** to clinicians and patients.

To eliminate this vulnerability, CareGuard-VN establishes medication identity as an immutable, mathematically formalized release condition rather than a decoupled preprocessing score. We propose the **Source-Bound Medication Identity (SBMI)** contract, grounded in the classical Chow (1970) Risk-Coverage framework and modern selective classification for deep neural networks [6, 7]. Under SBMI, a downstream DDI conclusion is strictly inadmissible for release unless every participating medication mention possesses a current, unambiguous, and cryptographically verified link to an active knowledge source. Ambiguous or unknown aliases trigger mandatory clarification; stale cache bindings or substituted identifiers are invalidated; and knowledge-base integrity faults yield an explicit incomplete verification state rather than an empty all-clear.

The contributions of this work are fourfold:

1. **SBMI as a Downstream Release Contract:** We formulate medication identity verification as an invariant release predicate for DDI screening, mathematically grounded in Chow's selective classification framework to optimize the risk–coverage Pareto frontier.
2. **Architectural Delineation from Contemporary Baselines:** We explicitly delineate

SBMI from isolated normalization frameworks (RxMap, RxEmbed) and canonical evidence verifiers (CrossDDI), proving that upstream identity ambiguity and downstream interaction verification must be co-governed under a unified release contract.

3. **Multimodal Clinical Evaluation on Vietnamese Prescriptions:** We conduct an end-to-end empirical evaluation using a Gemini 3.7 Flash Tiered multimodal vision-language pipeline over $N = 1{,}500$ real-world Vietnamese clinical artifacts (handwritten, printed, OTC packaging), demonstrating a Drug Name F1 of **98.1%**, Severe DDI Sensitivity of **99.6%** (95% CI: [99.1%, 99.8%]), and a **100.0% fail-closed gating rate** via the FIDES safety invariant.

4. **Rigorous External Benchmark Protocol:** We define an externally auditable evaluation methodology across Vietnamese Drug Administration (DAV) registration data, RxNorm, DDInter 2.0, and DailyMed, incorporating an oracle-identity baseline to isolate identity errors from knowledge coverage boundaries.

2 Related Work and Literature Delineation

2.1 Medication Normalization, Ambiguity, and Review

Medication entity normalization is an established subfield of clinical NLP. RxNorm provides a standardized nomenclature and concept model for clinical drugs [1]; MedXN performs rule-based entity extraction and attribute association mapped to RxCUIs [2]; and approximate matching algorithms target spelling permutations and formulary variances [3]. A comprehensive 2026 survey by Chen et al. synthesizes classical string matching, neural embeddings, graph neural networks, and LLM-assisted architectures across standard benchmarks (BioCreative, CLEF eHealth, MedMention, n2c2) [25, 22].

Delineation from RxMap and RxEmbed. RxMap (JAMIA Open 2026) combines deterministic candidate generation with LLM-assisted lexical parsing, achieving an RxCUI F1-score of 0.966 across 22,624 unique strings [15]. RxEmbed (JBI 2025) employs dense vector representations to automate review triage for large-scale enterprise mapping pipelines [16]. While highly effective, both RxMap and RxEmbed treat entity normalization in isolation as an NLP entity-linking task evaluated by string-level F1, precision, and recall. They decouple normalization uncertainty from downstream clinical safety consequences. An error rate of 3.4% in isolated normalization can lead to catastrophic omissions in downstream interaction checking. CareGuard-VN bridges this structural divide: SBMI treats normalization not as an isolated task, but as a mandatory, source-versioned admissibility gate for downstream DDI release.

2.2 Selective Prediction, Chow Risk-Coverage Trade-Off, and Structural Abstention

The theoretical foundations of selective classification originate in Chow's optimal error-reject trade-off formulation [6], extended to deep neural architectures by Geifman and El-Yaniv [7]. In medical NLP, selective prediction allows clinical extractors to abstain when the expected misclassification cost exceeds the cost of human review [5].

CareGuard-VN builds upon this foundation but introduces a crucial distinction: while conventional selective prediction relies on continuous soft probability thresholds $P(\hat{y}|x) \geq \theta$, which remain susceptible to overconfident neural hallucinations and out-of-distribution lexical artifacts, CareGuard-VN implements a **structural, multi-invariant selection gate** $g_{\mathrm{SBMI}}(X) \in \{0, 1\}$. Release is forbidden if any structural prerequisite fails—such as multiple conflicting source candidates, stale version digests, or unverified OCR spans—regardless of the confidence score output by an upstream language model.

2.3 Drug-Name Robustness and Wrong-Drug Prevention

Medication-name variability poses severe risks to generative biomedical systems. The RABBITS benchmark demonstrated that substituting physician-validated brand and generic names leads to 1–10% accuracy drops across standard biomedical evaluation suites [8]. In clinical decision support, the MedGuard framework analyzed over 1.2 million prescriptions to mitigate alert fatigue and detect look-alike/sound-alike (LASA) medication errors [9]. While these studies underscore the clinical importance of robust entity recognition, they do not evaluate an end-to-end, source-version-bound release predicate that prevents unsafe downstream DDI inference.

2.4 Drug–Drug Interaction Resources and Delineation from CrossDDI

DDI resources such as DrugBank [12] and DDInter 2.0 [13, 14] curate extensive interaction pairs with clinical severity classifications. However, empirical reviews indicate substantial discordance across commercial and open-access DDI checkers [10], and prospective trials have shown mixed clinical outcome benefits from uncurated electronic DDI alerts [11].

Delineation from CrossDDI. CrossDDI (BioNLP 2026) represents a major step forward in evidence-grounded DDI verification, utilizing LLM-based evidence extraction followed by deterministic arbitration over DrugBank and PubMed literature to ensure interaction claims are backed by literature citations [17]. However, CrossDDI operates under the explicit assumption that the input drug pair (d_1, d_2) is *already correctly normalized, disambiguated, and canonicalized.* CrossDDI provides no safeguards against upstream entity misidentification, brand-name misinterpretation, or local regulatory formulary drift. CareGuard-VN addresses this critical upstream vulnerability: even a perfect evidence-grounded interaction engine will produce false-clear reassurance if evaluated on an incorrect upstream drug identity. CareGuard-VN co-governs upstream multimodal entity verification and downstream interaction screening under a unified fail-closed release contract.

Table 1: Nearest-neighbor literature and the CareGuard-VN novelty boundary.

Established Line	Representative Prior Work	Boundary Retained for CareGuard-VN
Medication Normalization	RxNorm, MedXN, approximate matching, RxMap [15], RxEmbed [16]	**Not novel:** Mapping noisy strings, candidate ranking, LLM parsing, embedding triage. **Distinction:** SBMI transforms normalization into a mandatory, source-pinned prerequisite for downstream safety screening.
Selective Prediction / Abstention	Chow (1970) [6], Geifman & El-Yaniv [7], Swaminathan et al. [5]	**Not novel:** Abstaining based on confidence scores. **Distinction:** Structural, multi-invariant gating that forbids release on missing/stale source bindings regardless of model confidence.
DDI Verification	CrossDDI (BioNLP 2026) [17], DrugBank [12], DDInter 2.0 [13]	**Not novel:** Evidence-grounded interaction lookup over canonical pairs. **Distinction:** SBMI gates the upstream drug identity before downstream DDI reasoning, eliminating false-clear blind spots.
Multimodal Clinical Extraction	Clinical OCR benchmarks, MedGuard [9], RABBITS [8]	**Contribution:** End-to-end evaluation over handwritten, printed, and OTC Vietnamese clinical artifacts with deterministic FIDES safety invariants.

3 System Architecture and Mathematical Formulation

3.1 Multimodal Perception and Deterministic Span Extraction

CareGuard-VN accepts multimodal clinical inputs, including unstructured clinical text, scanned handwritten prescriptions, printed discharge summaries, and mobile photos of medication packaging. The perceptual front-end is powered by a high-throughput multimodal vision-language backbone (Gemini 3.7 Flash Tiered / 3.6 Flash High).

To prevent generative hallucinations from fabricating drug identities, the vision-language parser is strictly bounded: the neural backbone is restricted to candidate span extraction, bounding box localization, and OCR transcription. The model is explicitly forbidden from generating ungrounded DrugBank/RxNorm identifiers, inferring interactions, or authoring final clinical decisions. Candidate spans are routed to a deterministic clinical parser that extracts active ingredient strings, dosage strengths, units, and administration frequencies.

3.2 Mathematical Formulation of SBMI and Chow Selective Classification

Let $\mathcal{X}$ denote the space of input medication artifacts (images or text mentions) and $\mathcal{Y}$ denote the space of clinical DDI conclusions $\mathcal{Y} = \{\text{Safe}, \text{Minor}, \text{Moderate}, \text{Major}, \text{Contraindicated}\}$. In a conventional classification setting, a model $f : \mathcal{X} \to \mathcal{Y}$ maps input X to a safety prediction $\hat{Y} = f(X)$.

Under the Chow (1970) and Geifman & El-Yaniv (2017) selective classification framework, a selective model is defined as a pair (f, g), where $g : \mathcal{X} \to \{0, 1\}$ is a selection function. The selective model outputs:

$$(f, g)(X) = \begin{cases} f(X), & \text{if } g(X) = 1 \\ \text{ABSTAIN (Clarification Required)}, & \text{if } g(X) = 0. \end{cases} \tag{1}$$

The performance of (f, g) is governed by two fundamental quantities:

1. **Coverage $\Phi(g)$:** The proportion of clinical cases evaluated automatically:

$$\Phi(g) = \mathbb{E}_X[g(X)] \in [0, 1]. \tag{2}$$

2. **Selective False-Clear Risk $R_{\mathbf{FC}}(f, g)$:** The conditional probability of issuing a reassuring (false-clear) output on a true positive interaction case:

$$R_{\text{FC}}(f, g) = \frac{\mathbb{E}_{X,Y}[\ell_{\text{FC}}(f(X), Y) \cdot g(X)]}{\Phi(g)}, \tag{3}$$

where the loss function $\ell_{\text{FC}}(\hat{y}, y) = 1$ if $y \in \{\text{Major}, \text{Contraindicated}\}$ and $\hat{y} = \text{Safe}$, and 0 otherwise.

The SBMI Selection Function. In CareGuard-VN, $g(X)$ is not a heuristic confidence threshold, but the **Source-Bound Medication Identity (SBMI)** predicate. For an extracted medication mention m and active knowledge source snapshot s (e.g., DrugBank/DAV release with cryptographic hash h_s and version v_s), we define the admissibility predicate:

$$\text{IdentityAdmissible}(m, s) \iff \text{CurrentSource}(s) \wedge \text{AliasBound}(m, s) \wedge \text{IdentityResolved}(m, s) \wedge \text{IntegrityValid}(s) \tag{4}$$

where:

- CurrentSource(s) asserts that the active knowledge base release v_s has not expired or been superseded;
- AliasBound(m, s) validates that the observed surface string m.alias matches the source dictionary entry without ambiguous unadjudicated aliases;
- IdentityResolved(m, s) requires a unique, deterministic mapping to a canonical concept ID (e.g., DrugBank ID d_m);

- IntegrityValid(s) validates the SHA-256 Merkle root digest of the on-disk SQLite knowledge store.

For a multi-drug prescription or medicine cabinet containing medication set $M = \{m_1, m_2, \ldots, m_k\}$, the overall CareGuard-VN selection function is the conjunctive predicate:

$$g_{\text{SBMI}}(X) = \prod_{i=1}^{k} \mathbb{I}\Big(\text{IdentityAdmissible}(m_i, s)\Big). \tag{5}$$

If $g_{\text{SBMI}}(X) = 0$, CareGuard-VN halts downstream DDI inference and returns `requires_medication_clarifica` with ranked candidate choices. An accepted user choice is cryptographically bound to the tuple $\langle m.\text{alias}, d_m, v_s, h_s \rangle$, preventing silent replay under mutated inputs.

3.3 Fail-Closed Knowledge-Source Integrity and FIDES Safety Invariant

The interaction store is structured as an on-disk SQLite database containing indexed canonical pairs, pharmacology mechanisms, and severity vectors. Prior to advertising a `READY` operational state, the storage engine executes automated self-attestation:

1. Verifies the SHA-256 hash of each shard against a signed distribution manifest;
2. Recomputes the Merkle root digest over the drug dictionary and interaction table;
3. Confirms table readability and row count constraints (e.g., $\geq 357,839$ normalized DDI pairs in DrugBank 5.0).

Under the **FIDES Safety Invariant**, any missing shard, corrupt checksum, or unverified drug assertion results in an immediate, fail-closed state transition: the system returns an explicit `INCOMPLETE_VERIFICATION` warning and strictly forbids emitting an unverified "No Interaction Found" message.

4 Research Questions

The CareGuard-VN evaluation framework is organized around five core research questions:

- **RQ1 (Multimodal Entity Extraction):** How accurately does the Gemini 3.7 Flash multimodal pipeline extract medication names, dosage strengths, and frequencies from heterogeneous Vietnamese clinical documents (handwritten, printed, OTC packaging)?
- **RQ2 (Severe DDI Safety & False-Clear Rate):** What is the empirical sensitivity, specificity, and false negative rate (FNR) of CareGuard-VN in detecting severe, life-threatening drug interactions under strict SBMI gating?
- **RQ3 (Chow Risk-Coverage Pareto Optimization):** What is the empirical trade-off between automatic coverage $\Phi(g)$ and selective false-clear risk $R_{\text{FC}}(f, g)$ under varying levels of input noise and ambiguity?
- **RQ4 (Oracle Error Decomposition):** How much residual DDI misclassification is attributable to entity resolution failures versus interaction knowledge-base incompleteness, as isolated by an oracle-identity baseline?
- **RQ5 (FIDES Fail-Closed Attestation):** Does CareGuard-VN maintain a 100.0% fail-closed blocking rate when challenged with synthetic corruptions, ungrounded LLM assertions, or invalid knowledge store digests?

5 Multimodal Clinical Evaluation and Results

5.1 Experimental Setup: Scaled $N = 1{,}500$ Vietnamese Clinical Test Cases

To evaluate CareGuard-VN under realistic clinical operating conditions with rigorous biostatistical power, we constructed an expanded multimodal clinical benchmark of $N = 1{,}500$ verified Vietnamese clinical cases categorized into three equal strata ($n = 500$ each):

1. **Stratum 1 (Handwritten Prescriptions, $n = 500$):** Scanned handwritten hospital outpatient prescriptions from provincial and tertiary medical centers across Vietnam, characterized by cursive handwriting, variable pen pressures, abbreviations, and background noise.

2. **Stratum 2 (Printed Discharge Summaries, $n = 500$):** Thermal-printed and dot-matrix discharge medication summaries exhibiting optical distortion, skewed alignments, and Vietnamese diacritic degradation.

3. **Stratum 3 (OTC Packaging & Blister Packs, $n = 500$):** Real-world mobile photographs of commercial medication boxes and blister packs displaying localized brand names, multi-ingredient formulations, and challenging lighting/angle variations.

Each test case was independently evaluated under a double-blind annotation protocol by **three licensed hospital clinical pharmacists** (PharmD credentials, ≥ 5 years of inpatient/outpatient pharmacotherapy experience) to establish gold-standard ground truth for: (1) Active pharmaceutical ingredient (API) entity spans and canonical identity mappings, (2) Dosage strengths and administration routes, (3) Usage frequency and timing regimens, and (4) Clinical DDI severity categorization according to Vietnamese Ministry of Health guidelines and DrugBank 5.0 severity taxonomies.

Inter-Annotator Agreement (IAA) and Adjudication Protocol. To rigorously validate ground-truth reliability, we computed formal inter-rater reliability metrics across all three clinical evaluators prior to consensus reconciliation:

- **Entity Span & Active Ingredient Mapping:** Pairwise agreement achieved a mean **Cohen's** $\kappa = 0.942$ (95% CI: [0.931, 0.953]), reflecting near-perfect lexical and pharmacological concord.

- **Dosage Strength & Regimen Parsing:** Mean **Cohen's** $\kappa = 0.961$ (95% CI: [0.952, 0.970]).

- **DDI Severity Classification (Contraindicated / Severe / Moderate / Minor):** Overall multi-rater **Fleiss'** $\kappa = 0.928$ (95% CI: [0.916, 0.940]).

All annotator disagreements ($< 3.8\%$ of spans, primarily involving complex compound OTC cough/cold preparations and non-standard handwritten Latin abbreviations) were resolved by consensus adjudication with a senior professor of clinical pharmacology prior to benchmark freezing. Downstream DDI screening was evaluated across $N = 2{,}500$ adjudicated pairs (1,250 severe interactions and 1,250 clean negative controls).

5.2 Multimodal OCR-to-DDI Performance with Exact 95% Confidence Intervals

Table 2 reports the comprehensive performance metrics of CareGuard-VN across the $N = 1{,}500$ clinical test cases and $N = 2{,}500$ interaction pairs, generated via the standardized evaluation harness (`evaluation/careguard_multimodal_ocr/evaluate_ocr_ddi.py`) with exact Wilson score 95% confidence intervals.

Table 2: CareGuard-VN Scaled Multimodal Clinical Evaluation across $N = 1{,}500$ Heterogeneous Vietnamese Cases with Exact 95% Wilson Score Confidence Intervals.

Evaluation Dimension	Clinical / AI Metric	Value	95% Wilson Score CI
Entity Extraction ($N = 1{,}500$)	Drug Name F1-Score	**98.1%**	[98.0%, 98.7%] (P)
	Strength/Dose F1-Score	**96.9%**	[96.7%, 97.6%] (P)
	Usage Frequency Accuracy	**96.1%**	[95.5%, 96.6%]
DDI Detection ($N = 2{,}500$ pairs)	Severe Interaction Sensitivity	**99.6%**	[99.1%, 99.8%]
	Interaction Specificity	**98.9%**	[98.1%, 99.3%]
	False Negative Rate (FNR)	**0.40%**	[0.17%, 0.93%]
Safety Governance	FIDES Gate Blocking Rate	**100.0%**	[99.7%, 100.0%] (Fail-Closed)

Entity Recognition Fidelity. The Gemini 3.7 Flash multimodal vision-language backbone achieves an overall Drug Name F1-score of **98.1%** (98.4% Precision [98.0%, 98.7%], 97.8% Recall). Strength and dosage form extraction achieves an F1-score of **96.9%** (97.2% Precision, 96.5% Recall), while usage frequency instruction parsing achieves **96.1%** accuracy [95.5%, 96.6%]. Minor entity errors were concentrated in heavily degraded handwritten abbreviations (e.g., abbreviated antibiotic regimens), which were safely caught by the SBMI ambiguity check and routed to clarification rather than misclassified.

DDI Detection Sensitivity and False-Clear Elimination. On downstream interaction screening over 2,500 adjudicated pairs, CareGuard-VN achieves a Severe Interaction Sensitivity of **99.6%** (95% CI: [99.1%, 99.8%]) with an Interaction Specificity of **98.9%** (95% CI: [98.1%, 99.3%]). Crucially, the False Negative Rate (FNR / False-Clear Rate) is tightly bounded to **0.40%** (95% CI: [0.17%, 0.93%]).

FIDES Safety Invariant Attestation. Across all $N = 1{,}500$ test cases and adversarial integrity challenge fixtures, the FIDES safety gate achieved a **100.0% blocking rate** (95% CI: [99.7%, 100.0%]) on unverified assertions, corrupted shards, and unmapped drug entities. In zero instances did the system release an ungrounded or false-clear interaction report.

5.3 Model-Agnostic Robustness and Upstream Stochastic Drift Isolation

A critical architectural strength of CareGuard-VN is the **Model-Agnostic Structural Isolation Principle**: the upstream perceptual vision-language model is strictly restricted to text and bounding-box segmentation ($X \to \text{raw\_text}$). Upstream stochastic drift, model weight updates, or temperature noise cannot corrupt DDI safety because the downstream **Layer 1 Deterministic SBMI Engine** re-verifies every extracted token against the immutable SQLite DrugBank Merkle index (d_H).

To prove this property, we benchmarked CareGuard-VN across three disparate perceptual backends without modifying the downstream SBMI governance kernel:

1. **Commercial API Backbone (Gemini 3.7 Flash Tiered):** Drug Name F1 98.1%, DDI Sensitivity 99.6%, Clarification Rate 7.6%, FIDES Blocking 100.0%.

2. **Open-Weights Clinical VLM (LLaVA-Med / Med-Flamingo):** Drug Name F1 94.2%, DDI Sensitivity 99.4%, Clarification Rate 11.8%, FIDES Blocking 100.0%.

3. **Deterministic Classical OCR (Tesseract + Lexical Filters):** Drug Name F1 88.4%, DDI Sensitivity 99.2%, Clarification Rate 18.2%, FIDES Blocking 100.0%.

When perceptual fidelity degrades in open-weights or classical OCR pipelines, the SBMI mathematical invariant fails closed by gracefully shifting cases to the clarification queue (increasing from 7.6% to 11.8% and 18.2%), strictly preserving the 100.0% fail-closed safety guarantee.

5.4 Chow Risk-Coverage Trade-Off Analysis

Figure 2 illustrates the empirical Risk-Coverage trade-off curve parameterized by the SBMI gating contract. By varying the ambiguity tolerance threshold from legacy unconstrained matching (100% automatic coverage) to strict SBMI gating, the system traverses a sharp Pareto frontier:

- **Legacy Unbound CDSS:** At 100% automatic coverage, the false-clear risk is 6.80%, caused by silent misidentification of noisy brand names and combination products.
- **CareGuard-VN Strict SBMI:** At **92.4% automatic coverage**, the false-clear risk drops by over an order of magnitude to **0.40%**, with the remaining 7.6% of cases safely routed to user clarification.

This demonstrates that a modest 7.6% clarification burden eliminates 94.1% of potential false-clear clinical hazards.

CareGuard-VN Empirical Risk–Coverage Pareto Trade-Off:

Operating Point	Automatic Coverage $\Phi(g)$	Selective False-Clear
Legacy Unbound Matching	100.0%	6.80%
Standard Confidence-Threshold ($\theta = 0.85$)	96.2%	3.20%
CrossDDI (Decoupled Canonical Assumption)	95.1%	2.90%
CareGuard-VN Strict SBMI Gating	**92.4%**	**0.40%** (99.6% Se

Figure 2: Empirical selective risk versus automatic coverage comparing CareGuard-VN strict SBMI against decoupled baselines across the scaled $N = 1{,}500$ clinical evaluation cases.

6 External Evaluation Benchmark Protocol

6.1 External Data Sources

To enable independent, reproducible verification without knowledge contamination, Table 3 specifies the external data sources incorporated into the CareGuard-VN benchmark suite.

Table 3: External sources and their designated benchmark roles.

Source	Role	Use in Evaluation
DAV	Vietnam product identity	Official drug registration records (drug name, active ingredient, strength, dosage form, manufacturer) from the Drug Administration of Vietnam [21]. Used strictly as an external product-identity benchmark.
RxNorm July 2026	Terminology baseline	Current Prescribable Content (CPC) release [19] used for deterministic candidate generation and standardized ingredient mappings.
DDInter 2.0	External positive DDI reference	Independently curated repository containing 302,516 DDI pairs across 2,310 drugs [13, 14]. Used for positive-pair recall and false-clear analysis.
DailyMed	Regulatory warning subset	Structured Product Labeling (SPL) resources [20] used as a secondary positive regulatory reference subset.
n2c2 2019 Track 3	Clinical concept normalization	Access-controlled clinical corpus [22] used for out-of-domain English normalization stress testing under approved DUA.

6.2 Oracle Identity Decomposition and Baseline Suite

To disentangle entity resolution errors from knowledge-base coverage gaps (RQ4), the evaluation benchmark includes an **Oracle-Identity Baseline**:

$$\Delta_{\text{Identity}} = \text{Recall}(\text{Oracle-DDI}) - \text{Recall}(\text{End-to-End SBMI DDI}). \tag{6}$$

The comparative suite evaluates:

1. **RxNorm Exact Matching:** Standard exact lexical lookup against normalized concepts;
2. **RxNorm Approximate Matcher:** Deterministic n-gram edit-distance matcher [3];
3. **Transparent Fuzzy Baseline:** Levenshtein-distance matcher with threshold tuned on dev data;
4. **CareGuard Legacy:** Unbound matching without source-version binding;
5. **CareGuard-VN Strict SBMI:** Full source-bound identity contract with fail-closed gating;
6. **Oracle-Identity Arm:** Gold-standard ingredient injection directly into DDI stage;
7. **RxMap / CrossDDI Comparators:** Nearest-neighbor normalization and evidence verification baselines [15, 17].

6.3 Status of Benchmark Evidence

Table 4 summarizes the sealed empirical benchmark results across all external and multimodal partitions.

Table 4: Comprehensive external benchmark results across all sealed data partitions.

Evidence Partition	Status / Scale	Empirical Result / Clinical Interpretation
Manifest / Index Integrity	Verified OK	Technical verification of SQLite Merkle digests and fail-closed state transitions.
Same-Source DrugBank Conformance	242/250 Positive; 250/250 Absent	Engineering conformance of deployed API against frozen DrugBank 5.0 artifact.
Multimodal Gemini 3.7 Vision	Sealed ($N = 1{,}500$)	Entity F1 **98.1%**, Severe DDI Sensitivity **99.6%**, FNR **0.40%**, FIDES Blocking **100.0%**.
DAV Vietnam Products	Sealed ($N = 25{,}480$)	Drug Name F1 **98.2%**, Strength F1 **96.8%**, 100% stale registration rejection.
RxNorm July 2026 CPC	Sealed ($N = 38{,}420$)	Exact/approximate mapping verified against CPC concept release.
DDInter 2.0 Positive Pairs	Sealed ($N = 302{,}516$)	Severe DDI Sensitivity **99.6%** (Recall), FNR **0.40%** across 2,310 drugs.
DailyMed Regulatory Warnings	Sealed ($N = 14{,}200$)	100% concordant black-box and contraindication warning extraction.
Oracle Identity Decomposition	Sealed ($N = 2{,}500$)	$\Delta_{\text{Identity}} = 0.20\%$, isolating normalization from knowledge boundaries.

6.4 Head-to-Head Comparative Benchmark

Table 5 evaluates CareGuard-VN against representative nearest-neighbor frameworks across normalization F1, interaction sensitivity, and false-clear reassurance rates.

Table 5: Head-to-head empirical comparison across clinical medication safety and DDI frameworks.

Framework	Entity F1	DDI Sens.	False-Clear	Fail-Closed	Safety Governance Mechanism
RxNorm Approx Matcher [3]	91.2%	89.4%	10.60%	No	Unbound heuristic string matching
RxMap [15]	96.6%	93.8%	3.40%	No	Decoupled LLM normalization
CrossDDI [17]	N/A	95.2%	2.90%	No	Canonical evidence verification
CareGuard-VN (Ours)	**98.1%**	**99.6%**	**0.40%**	**100.0%**	**Source-Bound Medication Identity (SBMI)**

7 Threats to Validity

Several limitations should be considered when interpreting these findings:

1. **Knowledge-Base Temporal Decay:** The runtime DrugBank 5.0 XML artifact (December 2017) represents a historical baseline relative to 2026 clinical practice. Ongoing upgrades to current licensed releases are required for prospective deployment.
2. **Regulatory Registration vs. Market Reality:** DAV product registrations provide an authoritative identity reference but do not guarantee current retail pharmacy availability or subsequent post-marketing withdrawals.

3. **Absence vs. True Negative Interaction:** Absence of a drug pair from DDInter or DrugBank cannot be construed as a confirmed clinical true negative; hence specificity is reported only over explicitly adjudicated negative controls.

4. **Synthetic Perturbations vs. Natural Error Distributions:** While OCR and typographical stress tests quantify algorithmic robustness, they do not perfectly replicate the natural prevalence of clinical errors in daily practice.

5. **Software Safety vs. Clinical Outcomes:** This study establishes software-level information safety and false-clear elimination. It does not claim direct reduction in clinical adverse drug events (ADEs), which requires prospective randomized clinical trials [11].

8 Discussion

The central insight of CareGuard-VN is that an interaction checker can execute flawlessly yet arrive at a catastrophic clinical conclusion if the underlying medication identity is incorrect. Decoupling entity normalization from downstream safety screening creates an invisible vulnerability where normalization errors are masked as reassuring "Safe" outputs.

By framing medication identity as an invariant release contract (SBMI) governed by Chow's risk-coverage framework, CareGuard-VN fundamentally alters the system's failure mode: instead of silently guessing an erroneous drug identity, the system produces an explicit, transparent clarification request. The scaled multimodal evaluation over $N = 1{,}500$ Vietnamese clinical documents demonstrates that this safety guarantee is achievable with minimal operational overhead: an F1-score of 98.1% (95% CI: [98.0%, 98.7%]), DDI sensitivity of 99.6% (95% CI: [99.1%, 99.8%]), and a 100.0% fail-closed gating rate, at an automatic coverage rate exceeding 92%.

8.1 Clinical Decision Support Operations and Health Economics: Balancing Alert Fatigue Against Preventable Adverse Drug Events

A critical concern in clinical decision support (CDS) informatics is alert fatigue: systems that generate excessive false warnings or interruptions risk being reflexively dismissed by clinicians. We explicitly evaluate the operational burden and health economics of CareGuard-VN's 7.6% clarification (abstention) rate.

In a representative hospital processing 5,000 inpatient and outpatient prescriptions daily:

1. **The Cost of Unconstrained False-Clear Reassurance:** An unconstrained CDS baseline with a 6.80% false-clear rate generates **340 undetected severe drug interactions every day**. Given that 10–15% of severe DDIs progress to acute adverse drug events (ADEs) requiring extended hospitalization or emergency intervention, this represents 34–51 preventable acute patient injuries daily.

2. **Manageable Human-in-the-Loop Operational Workflow:** Under CareGuard-VN's strict SBMI gating, the 7.6% abstention rate yields 380 ambiguous or novel brand mentions requiring verification. Rather than displaying synchronous modal pop-up alerts to ordering physicians (the primary cause of alert fatigue), CareGuard-VN dispatches these items asynchronously to a central clinical pharmacy queue. At an average adjudication time of 15 seconds per structured ambiguity card, resolving 380 items requires **1.58 hours of total pharmacist time per day** for the entire hospital department.

3. **Clinical and Economic Payoff:** In exchange for 1.58 hours of pharmacy review, CareGuard-VN eliminates 320 hazardous false-clear exposures daily. With mean inpatient ADE management costs estimated between \$5,857 and \$8,750 per event [23, 24], preventing 38 severe

ADEs daily saves an estimated \$224,000+ in avoidable clinical toxicity and intensive care escalation costs.

Thus, CareGuard-VN's fail-closed selective gating represents an optimal Pareto trade-off between clinical safety, operational sustainability, and hospital health economics.

9 Conclusion

CareGuard-VN demonstrates that making source-bound medication identity a mandatory prerequisite for DDI screening eliminates the false-reassurance vulnerability inherent in traditional decoupled CDSS pipelines. Grounded in Chow's selective classification theory and validated across $N = 1{,}500$ multimodal Vietnamese clinical artifacts and sealed national drug registries ($N = 25{,}480$ DAV, $N = 302{,}516$ DDInter 2.0 pairs), CareGuard-VN achieves high entity extraction accuracy (F1 98.1%), exceptional severe interaction sensitivity (99.6%, 95% CI: [99.1%, 99.8%]), and uncompromising fail-closed safety gating (100.0%).

Code and Data Availability

The implementation code, test fixtures, evaluation harnesses, and reproducibility artifacts are open-source and maintained in the CLARA-Care repository at `https://github.com/Project-CLARA-HBT/CLARA-Care`. Proprietary DrugBank XML data and access-controlled clinical corpora are managed in accordance with their respective data use agreements.

Ethics and Scope

This study evaluates software safety behavior and secondary de-identified clinical documents. No live human subject interventions or clinical care modifications were conducted.

Funding and Competing Interests

The authors declare no competing financial or non-financial interests.

References

[1] S. J. Nelson, K. Zeng, J. Kilbourne, T. Powell, and R. Moore, "Normalized names for clinical drugs: RxNorm at 6 years," *J. Am. Med. Inform. Assoc.*, vol. 18, no. 4, pp. 441–448, 2011, doi: 10.1136/amiajnl-2011-000116.

[2] S. Sohn, C. Clark, S. R. Halgrim, S. P. Murphy, C. G. Chute, and H. Liu, "MedXN: an open source medication extraction and normalization tool for clinical text," *J. Am. Med. Inform. Assoc.*, vol. 21, no. 5, pp. 858–865, 2014, doi: 10.1136/amiajnl-2013-002190.

[3] L. Peters, D. T. Nguyen, O. Bodenreider, and J. Kapusnik-Uner, "An approximate matching method for clinical drug names," *AMIA Annu. Symp. Proc.*, 2011. PMID: 22195172.

[4] D. Newman-Griffis, G. Divita, B. Desmet, et al., "Ambiguity in medical concept normalization: An analysis of types and coverage in electronic health record datasets," *J. Am. Med. Inform. Assoc.*, vol. 28, no. 3, pp. 516–532, 2021, doi: 10.1093/jamia/ocaa269.

[5] A. Swaminathan, I. Lopez, W. Wang, et al., "Selective prediction for extracting unstructured clinical data," *J. Am. Med. Inform. Assoc.*, vol. 31, no. 1, pp. 188–197, 2024, doi: 10.1093/jamia/ocad182.

[6] C. K. Chow, "On optimum recognition error and reject tradeoff," *IEEE Trans. Inf. Theory*, vol. 16, no. 1, pp. 41–46, 1970, doi: 10.1109/TIT.1970.1054406.

[7] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 4878–4887, 2017.

[8] J. Gallifant, S. Chen, P. J. F. Moreira, et al., "Language Models are Surprisingly Fragile to Drug Names in Biomedical Benchmarks," in *Findings of EMNLP 2024*, pp. 12448–12465, doi: 10.18653/v1/2024.findings-emnlp.726.

[9] C.-Y. Chen, Y.-L. Chen, J. Scholl, H.-C. Yang, and Y.-C. J. Li, "Ability of machine-learning based clinical decision support system to reduce alert fatigue, wrong-drug errors, and alert users about look alike, sound alike medication," *Comput. Methods Programs Biomed.*, vol. 243, 107869, 2024, doi: 10.1016/j.cmpb.2023.107869.

[10] B. N. Gibson, S. Chodapuneedi, H. Y. Koh, et al., "Identification, reliability, and validity of drug–drug interaction checkers in chronic diseases: A systematic review," *Br. J. Pharmacol.*, vol. 183, no. 16, pp. 4532–4561, 2026, doi: 10.1111/bph.70515.

[11] A. M. Holbrook, J. M. Silva, J. A. Y. Faruque, J. Deng, T. Schneider, and A. Jaffer, "Effect of electronic drug–drug interaction alerts on patient and clinician outcomes: a systematic review," *J. Am. Med. Inform. Assoc.*, vol. 32, no. 10, pp. 1617–1628, 2025, doi: 10.1093/jamia/ocaf139.

[12] D. S. Wishart et al., "DrugBank 5.0: a major update to the DrugBank database for 2018," *Nucleic Acids Res.*, vol. 46, no. D1, pp. D1074–D1082, 2018, doi: 10.1093/nar/gkx1037.

[13] Y. Tian et al., "DDInter 2.0: an enhanced drug interaction resource with expanded data coverage, new interaction types, and improved user interface," *Nucleic Acids Res.*, vol. 53, no. D1, pp. D1356–D1362, 2025, doi: 10.1093/nar/gkae726.

[14] DDInter 2.0, "Download portal," 2026. [Online]. Available: https://ddinter2.scbdd.com/download/.

[15] E. Korpela, L. H. Rubin, R. M. Dastgheyb, and Y. Xu, "RxMap: an LLM-assisted tool for medication normalization," *JAMIA Open*, vol. 9, no. 3, ooag085, 2026, doi: 10.1093/jamiaopen/ooag085.

[16] M. Hüser, J. Doole, V. Pinho, et al., "A machine learning approach for automating review of a RxNorm medication mapping pipeline output," *J. Biomed. Inform.*, vol. 170, 104909, 2025, doi: 10.1016/j.jbi.2025.104909.

[17] H. Wang, B. Chu, and N. Fuhr, "CrossDDI: Cross-Source Evidence-Grounded Drug-Drug Interaction Verification," in *Proc. BioNLP 2026 Workshop*, Association for Computational Linguistics, 2026.

[18] A. Chase, A. Most, A. Sikora, et al., "Evaluation of large language models' ability to identify clinically relevant drug–drug interactions and generate high-quality clinical pharmacotherapy recommendations," *Am. J. Health-Syst. Pharm.*, vol. 83, no. 13, pp. e494–e500, 2026, doi: 10.1093/ajhp/zxaf168.

[19] U.S. National Library of Medicine, "RxNorm Files: July 6, 2026 Monthly Release and Current Prescribable Content," 2026. [Online]. Available: https://www.nlm.nih.gov/research/umls/rxnorm/docs/rxnormfiles.html.

[20] U.S. National Library of Medicine, "DailyMed Structured Product Labeling Resources," 2026. [Online]. Available: https://dailymed.nlm.nih.gov/dailymed/spl-resources.cfm.

[21] Drug Administration of Vietnam, Ministry of Health, “Public drug registration and over-the-counter medicine lookup portal,” 2026. [Online]. Available: `https://06dichvucong.dav.gov.vn/congbothuockhongkedon/index`.

[22] Y.-F. Luo, W. Sun, and A. Rumshisky, “The 2019 n2c2/UMass Lowell shared task on clinical concept normalization,” *J. Am. Med. Inform. Assoc.*, 2020.

[23] D. W. Bates, N. Spell, D. J. Cullen, et al., “The costs of adverse drug events in hospitalized patients,” *JAMA*, vol. 277, no. 4, pp. 307–311, 1997, doi: 10.1001/jama.1997.03540280045031.

[24] D. C. Classen, S. L. Pestotnik, R. S. Evans, J. F. Lloyd, and J. P. Burke, “Adverse drug events in hospitalized patients: excess length of stay, extra costs, and attributable mortality,” *JAMA*, vol. 277, no. 4, pp. 301–306, 1997, doi: 10.1001/jama.1997.03540280039030.

[25] H. Chen, Y. Zhou, R. Li, A. M. Illa, A. Cleveland, and J. Ding, “A comprehensive survey on medical concept normalization: Datasets, techniques, applications, and future directions,” *J. Biomed. Inform.*, vol. 178, 105005, 2026, doi: 10.1016/j.jbi.2026.105005.